Collaborative Disassembly for Multiple Users in Virtual Reality

Ziteng Wang, Jian Wu, Sichun Huang, Peike Wang, Peng Zhang, Min Zhang, Lili Wang

***Abstract*—Disassembly Sequence Planning (DSP) is critical in product lifecycle management but computationally demanding, with traditional and machine learning approaches facing challenges in scalability and lacking the real-time responsiveness required for interactive collaboration in virtual reality (VR). We present Dynamic Parallel Disassembly and Tasking (DPDT), a novel framework tailored for immersive collaboration. DPDT integrates a spatial-temporal feature-based heuristic to accelerate search prioritization and employs a dynamic reactivation strategy to resolve inter-part dependencies during parallel execution. Furthermore, a proximity-aware task assignment algorithm optimizes the translation of these plans into efficient multi-user instructions. We conduct extensive evaluations on a large-scale dataset, showing that DPDT significantly outperforms state-of-the-art baselines. Specifically, for complex assemblies, our method achieves a computational speedup of up to 4.7 times while improving planning reliability by approximately 9.3 percentage points. These algorithmic gains translate directly into operational efficiency: user studies in VR confirm that DPDT significantly reduces task completion time by up to 56.5% and physical effort by over 50%, validating its potential for effective collaborative maintenance training.**

***Index Terms*—Virtual Reality, Disassembly Sequence Planning, Collaborative Disassembly, Human-Computer Interaction.**

I. Introduction

Disassembly Sequence Planning (DSP) is pivotal in various stages of a product's life cycle, including manufacturing, maintenance, repair, and recycling [1]. The inherent combinatorial complexity, where the number of possible sequences can explode exponentially with the number of components, makes these planning tasks exceptionally challenging [2].

Historically, addressing these challenges has involved various approaches, each with its own set of limitations. Searching and optimization based algorithms [3], [4], [5], while capable of finding optimal or near-optimal solutions, often grapple with extremely long computation times, particularly for complex assemblies. Moreover, these methods are frequently designed or tuned for specific object geometries or types, hindering their ability to be rapidly adapted or scaled to new, unseen models. This lack of generalization capability and speed poses a significant bottleneck in dynamic or rapidly changing operational environments. More recently, machine learning techniques have been applied to ASP/DSP, offering the potential to learn from data [6], [7], [8]. However, these methods often exhibit lower prediction success rates for individual disassembly steps. Consequently, to ensure a physically valid and correct disassembly sequence, multiple iterations of physical simulation are typically required for verification. If a predicted step fails in simulation, the system must backtrack and retry, leading to a cumulative increase in overall planning time and computational overhead. Moreover, existing methods do not yet support multi-user collaborative disassembly, and the unpredictability of their predictions often necessitates computationally expensive simulation verifications that hinder the real-time responsiveness required for virtual reality disassembly applications.

To address these challenges, we propose the Dynamic Parallel Disassembly and Tasking (DPDT) framework, a comprehensive pipeline designed to enable efficient multi-user collaborative disassembly in virtual environments. First, we develop a spatial-temporal feature-based disassembly priority prediction mechanism. By synthesizing geometric features, such as visibility and center distance, with dynamic temporal data from attempt history, this module effectively ranks candidate parts to accelerate the search for valid sequences. Second, we introduce a dynamic reactivation-driven parallel disassembly planning algorithm. This core component identifies sets of parts that can be removed concurrently and is enhanced by a novel dynamic reactivation strategy, which intelligently resolves emergent inter-part dependencies by updating candidate queues in real-time. Finally, to bridge the gap between algorithmic planning and human collaboration, we propose a proximity-aware task assignment strategy. This method transforms the parallel plan into interactive guidance, optimizing task distribution among multiple users by minimizing virtual travel distance. Extensive experiments demonstrate that DPDT achieves significant improvements in both computational speed and planning reliability over state-of-the-art methods. Furthermore, a user study confirms that our framework substantially reduces task completion time and physical workload in collaborative VR scenarios. Fig. 1 shows the workflow of our dynamic parallel disassembly and tasking method.

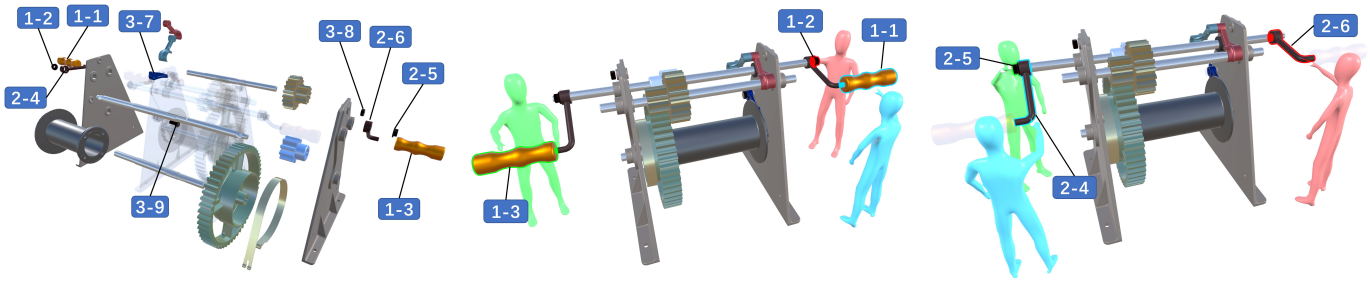
In summary, our contributions are as follows:

- We propose a pipeline for multiple users to collaboratively disassemble objects interactively in a virtual environment.
- We develop a spatial-temporal feature-based disassembly-priority prediction mechanism that increases the success rate of disassembly attempts and accelerates the planning process.
- We introduce a dynamic reactivation-driven parallel disas-

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(a) **(b)** **(c)**

Fig. 1: The workflow of our Dynamic Parallel Disassembly and Tasking (DPDT) framework, illustrating the process from planning to guided collaborative execution on a winch assembly. **(a)** The complete parallel disassembly sequence generated by our planning algorithm, where each part is labeled with its `round-id` to indicate its removal order. **(b)** The first round of disassembly in virtual reality: the system highlights and assigns the initial set of removable parts (1-1, 1-2, and 1-3) to users based on proximity to minimize physical effort. **(c)** A subsequent round of the task: once the parts from the previous round are removed, the system dynamically guides users to the next set of available parts (2-4, 2-5, and 2-6), ensuring a continuous and efficient collaborative workflow.

sembly planning method that adaptively updates features based on real-time disassembly feedback.

- We conducted a user study to evaluate the system's efficiency and user workload, demonstrating the effectiveness of our approach.

The remainder of this paper is organized as follows: Section II reviews related work in disassembly sequence planning, machine learning approaches, path planning, parallel disassembly, and collaborative task assignment. Section III details our proposed DPDT framework, including its part prioritization, planning, and task assignment algorithms. Section IV describes the experimental setup, metrics, results, and ablation studies for the planning component of DPDT. Section V presents the user study conducted to evaluate the complete DPDT framework in a VR environment. Finally, Section VI concludes the paper, outlines limitations and avenues for future research.

II. Related Work

In this section, we provide a brief review of research on searching and optimization methods, machine learning methods, the disassembling path-planning method, physical simulation, and parallel Disassembly Sequence Planning. We recommend that the readers refer to the reviews [1], [3], [6].

A. Searching and Optimization Methods

Assembly Sequence Planning (ASP) and its inverse, Disassembly Sequence Planning (DSP), are fundamental problems in product lifecycle management, impacting manufacturing, maintenance, and recycling [1]. These planning tasks are computationally challenging due to the combinatorial explosion of possible sequences, which can grow exponentially with the number of components [2]. Early research often relied on graph-based representations, such as AND/OR graphs [9], to model feasible sequences. However, constructing these graphs typically requires predefined geometric and precedence constraints, often derived from CAD models or annotated by domain experts [10], [11]. Once feasible sequences are identified, various optimization techniques, including genetic algorithms and heuristic search, have been employed to find optimal sequences based on criteria such as assembly time, cost, or stability [12], [4], [13]. To manage the inherent complexity, the "assembly-by-disassembly" strategy is widely adopted [14]. This approach leverages the typically more constrained nature of disassembly movements compared to assembly, thereby reducing the search space. For assemblies involving only rigid parts, a direct bijection often exists between disassembly and assembly sequences, allowing an assembly plan to be derived by reversing a valid disassembly sequence [3]. Our DPDT framework falls within this search-based paradigm but distinguishes itself by replacing computationally intensive global optimization with a dynamic, heuristic-driven strategy. Instead of relying on static precedence graphs, we utilize a spatial-temporal prioritization mechanism to guide the search, ensuring the computational efficiency and responsiveness required for interactive collaborative environments.

B. Machine Learning Methods

Recent advances in Machine Learning (ML) have opened new avenues for tackling ASP/DSP, as surveyed by Gerhard et al. [6]. Reinforcement Learning (RL) agents can learn optimal (dis)assembly policies through interaction within simulated environments, guided by carefully designed reward functions that encode task objectives [15], [16]. Graph Neural Networks (GNNs) are particularly well-suited for learning from the relational structure inherent in assemblies, enabling predictions of part removability or assembly priority based on graph representations encoding geometry and connectivity [7]. Furthermore, Transformer architectures, leveraging self-attention mechanisms, have been adapted to model long-range dependencies within graph-structured sequence planning tasks [17]. While these methods can achieve high performance and discover non-obvious strategies, they often require large, curated datasets for training, and their generalization to novel, out-of-distribution geometries can be a challenge. Moreover, the significant computational cost of training and the "black box" nature of some models can be prohibitive in certain industrial contexts. In contrast, our heuristic-based approach is data-free, computationally lightweight for inference, and

fully interpretable, making it robust for generic, unseen models without prior training. Physical simulations often play a crucial role in these ML approaches, serving not only for validation but also as environments for RL training or for generating synthetic data [18].

C. Disassembly Path Planning and Physical Simulation Methods

Beyond determining the sequence, finding a collision-free path for each disassembly or assembly step, known as Disassembly Path Planning (DPP) or Assembly Path Planning (APP), is crucial. While early work involved geometric reasoning [19], general-purpose sampling-based motion planning algorithms like Probabilistic Roadmaps (PRM) [20] and Rapidly-exploring Random Trees (RRT) [21] became prevalent. However, these methods can struggle in the highly constrained configuration spaces typical of (dis)assembly, often failing in situations involving narrow passages or complex contact interactions [22]. Physics-based motion planning [23] presents a compelling alternative, directly simulating physical forces and interactions to explore feasible paths. This is particularly advantageous for disassembly, where intricate contact states and non-linear escape trajectories are common. Recent work, such as Assemble Them All (ATA) by Tian et al. [5], has successfully employed the differentiable physical simulation **DiffRedMax** [24], [25], enhanced with Signed Distance Fields (SDF) for accurate and efficient contact resolution within an assembly-by-disassembly framework. This physics-centric approach, capable of discovering complex paths, was further refined with heuristics and learning-based guidance in ASAP [8]. Our DPDT framework utilizes the same simulation engine, DiffRedMax, integrating it into the disassembly attempt function to rigorously verify the physical feasibility of every planned step.

D. Parallel Disassembly Sequence Planning Methods

For complex products with many components, sequential disassembly can be prohibitively time-consuming. Parallel Disassembly Sequence Planning (PDSP) aims to accelerate the process by identifying sets of components that can be removed concurrently without interference. Research in this area has explored metaheuristic optimization algorithms, for example, Ant Colony Optimization (ACO) [26] and Particle Swarm Optimization (PSO) [27], to find efficient asynchronous parallel plans, often considering task dependencies and resource constraints. Our parallel DSP algorithm (Algorithm 2) within DPDT contributes to this area by specifically targeting multi-agent collaboration. Crucially, it introduces our dynamic reactivation strategy designed to handle dependencies that only become apparent during simultaneous disassembly attempts—a scenario particularly relevant in potentially unpredictable collaborative environments such as VR.

III. Method

This section details our DPDT framework, designed to enable efficient and ergonomic multi-user collaborative disassembly in Virtual Reality, as illustrated in Fig. 2. The pipeline consists of three sequential steps. The first step is spatial-temporal feature-based disassembly priority prediction (Section III-A), which synthesizes geometric and historical data to prioritize candidate parts, thereby accelerating the search for feasible sequences. The second step is dynamic reactivation-driven parallel disassembly planning (Section III-B), developed to maximize concurrent operations by identifying parallelizable parts and resolving inter-part dependencies through a dynamic update strategy. The third step is proximity-aware task assignment (Section III-C), implemented to optimize the collaborative workflow by minimizing user physical effort and translating the plan into interactive guidance.

A. Spatial-Temporal Feature-Based Disassembly Priority Prediction

The efficacy of any disassembly planner hinges on its ability to intelligently identify and prioritize viable candidate parts. We propose a spatial-temporal aware prioritization method that evaluates candidates by fusing spatial features with dynamic information derived from temporal features. This hybrid approach provides a more comprehensive and adaptive measure of a part's removability at any given stage.

1) Spatial Features: Spatial features quantify the geometric state of a part in relation to the rest of the assembly at a single point in time, including visibility metrics, center distance, and contacting parts.

- **Visibility (VAR, MVAR, SVAR):** Visibility serves as a strong indicator of a part's accessibility and lack of geometric occlusion. To compute it, we render the assembly from six orthogonal viewpoints $(+x, -x, +y, -y, +z, -z)$ to generate depth maps. For each viewpoint, the Visibility Area Ratio (*VAR*) of a part is defined as the ratio of its visible projected area to its total projected area. The total projected area is determined by counting the pixels corresponding to the part in its depth map, which are identified by having a depth value less than a maximum threshold, `maxD`, that represents the background. From the six *VAR* values, we derive two key metrics:
  - **Maximum Visibility Area Ratio (*MVAR*):** This is the highest value among the six *VAR*s. A high *MVAR* suggests the existence of at least one clear, non-occluded direction for removal, making it a primary criterion for identifying promising candidates.
  - **Sum of Visibility Area Ratios (*SVAR*):** This represents the part's overall exposedness from all directions. It serves as a secondary, tie-breaking feature.
- **Center Distance (CD):** This is the Euclidean distance between the geometric center of the candidate part and the centroid of the entire assembly. This feature is predicated on the heuristic that components located at the geometric periphery typically exhibit fewer spatial constraints and higher accessibility probability.
- **Contacting Parts (*contPs*):** While not a direct sorting feature, the set of contacting parts is critical for the temporal logic of the planner. When a disassembly attempt fails, the physics engine reports the specific parts that obstructed the removal. This contact information is

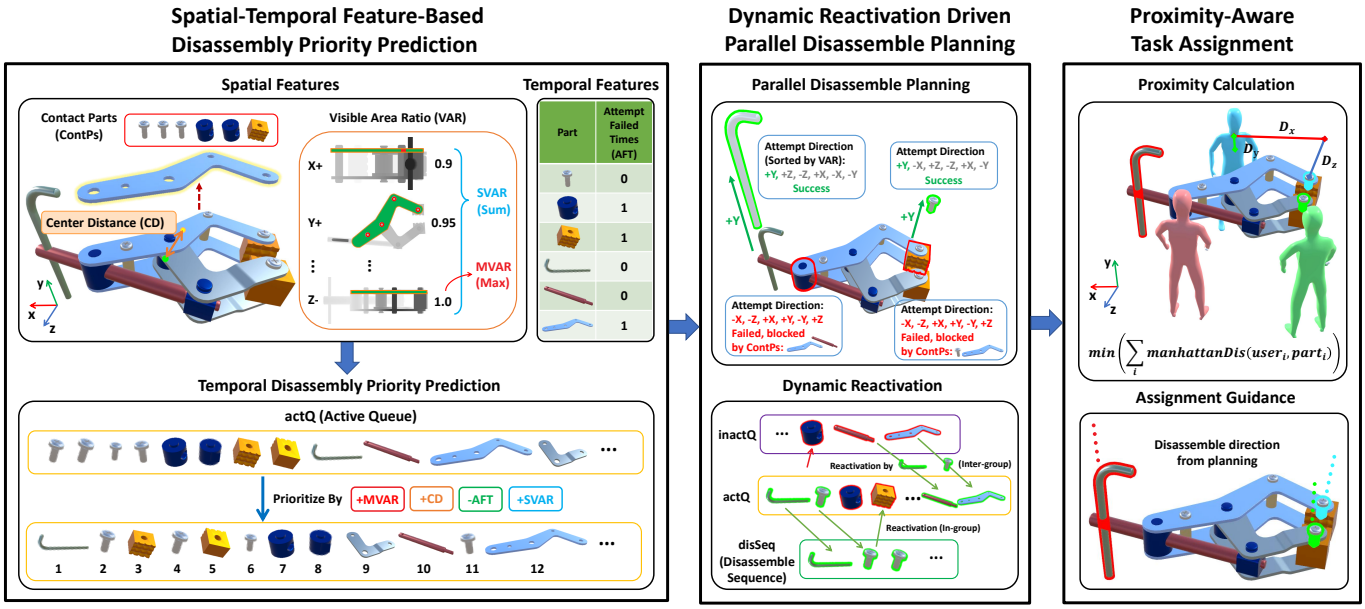


Fig. 2: The overall pipeline of our Dynamic Parallel Disassembly and Tasking (DPDT) framework. The process commences with Spatial and Temporal Disassembly Priority Prediction, where the system analyzes the assembly to compute spatial features, such as the Visibility Area Ratio and Center Distance, along with temporal data, including the number of Attempt Failed Times for each part. These metrics are synthesized to rank candidates in the active queue $actQ$, ensuring the most promising parts are addressed first. This prioritized queue then drives the Dynamic Reactivation Driven Parallel Disassembly Planning stage. Here, the algorithm attempts to remove multiple components concurrently and features a novel dynamic reactivation mechanism that intelligently resolves inter-part dependencies by promoting previously blocked parts from the inactive queue $inactQ$ back to the $actQ$ as obstructions are cleared. Finally, in the Proximity-Aware Task Assignment module, the successfully planned disassembly steps are distributed as tasks to users. This assignment optimizes the collaborative workflow by minimizing the total Manhattan distance between users and their assigned parts, and provides clear visual instructions through part highlighting and rendered disassembly paths to guide users effectively in the virtual environment.

stored in the *contPs* variable and is subsequently used by the dynamic reactivation mechanism to determine if a previously blocked part has become removable.

2) Temporal Features: Temporal features introduce a memory-based, adaptive component to the planning process, allowing the algorithm to learn from its own search history and avoid redundant computations. Specifically, we utilize the number of failed attempts to track historical interactions.

- **Attempt Failed Times (AFT):** This is a simple counter that increments each time a physics-based disassembly attempt `AtmptDis` fails for a specific part. The AFT acts as a penalty, dynamically lowering a part's priority after repeated failures. This mechanism prevents the planner from getting stuck in a loop, repeatedly trying to remove a part that is currently constrained, effectively implementing a penalty-based heuristic to balance the breadth and depth of search.

3) Spatial-Temporal Aware Prioritization: The input to this prioritization method is the current active queue ($actQ$) containing candidate parts. The output is the re-ordered $actQ$ sorted by removability likelihood, along with the computed feature set $STInfo$. The detailed implementation is presented in Algorithm 1. The process operates in two distinct phases: feature computation and queue sorting.

a) Feature Computation: First, the system iterates through all candidates to compute spatial metrics. To obtain visibility information, we employ a hardware-accelerated rendering approach within the nested loops (lines 2-3). The assembly is rendered from six orthogonal viewpoints $(+x,-x,+y,-y,+z,-z)$ to generate depth maps for both the target part and the whole assembly (lines 4-5). For a specific part p and viewpoint v, the Visibility Area Ratio (VAR) is computed (line 6) as:

$$VAR_{p,v} = \frac{\sum \mathbb{I}(D_{p,v} = D_{total,v})}{\sum \mathbb{I}(D_{p,v} < \texttt{maxD})} \tag{1}$$

where $D_{p,v}$ is the depth map of part p, $D_{total,v}$ is the depth map of the entire assembly, and $\mathbb{I}$ is the indicator function. Pixels with depth values less than the background threshold `maxD` constitute the projected area. Based on these views, $MVAR$ and $SVAR$ are derived via maximization and summation operations, respectively (lines 7-8). Simultaneously, the Center Distance (CD) is computed as the Euclidean distance between the part's bounding box center and the assembly's geometric centroid (line 9).

b) Sorting Strategy: Once the features are updated, the active candidate queue ($actQ$) is sorted in an order based on a hierarchical lexicographical key: $(-MVAR, -CD, AFT, -SVAR)$, where negative signs indicate

descending order (line 10). This composite key enforces the following prioritization logic:

1) $-MVAR$: Parts with the highest maximum visibility are prioritized first, as a high $MVAR$ serves as a robust geometric proxy for the existence of an unobstructed removal path.
2) $-CD$: Among parts with similar visibility, those located farther from the assembly center on outer layers are preferred, consistent with a periphery-to-core disassembly strategy.
3) AFT: To mitigate the risk of local optima, parts with fewer failed attempts are prioritized. This mechanism effectively penalizes candidates that are geometrically exposed but physically constrained, redirecting the search toward unexplored configuration spaces.
4) $-SVAR$: As a final tie-breaker, the component exhibiting greater global spatial exposure is selected.

By integrating precise geometric computation with this adaptive sorting strategy, our method ensures that the most promising candidates are consistently evaluated first, significantly reducing the search space.

Algorithm 1: Spatial-Temporal Aware Prioritization(STAwPri)

1 **Input**: Active queue $actQ$ **Output**: Spatial-Temporal aware prioritized active queue $actQ$, $STInfo$
2 **for** *part in* $actQ$ **do**
3 **for** *dir in* $[x,-x,y,-y,z,-z]$ **do**
4 $DMpart \leftarrow$ `GetDM`(dir, $part$)
5 $DMPList \leftarrow$ `GetDM`(dir, $PList$)
6 $STInfo[part].VAR[dir] \leftarrow$ `Sum`($DMpart = DMPList$) / `Sum`($DMpart < maxD$)
7 $STInfo[part].MVAR \leftarrow$ `Max`($STInfo[part].VAR$)
8 $STInfo[part].SVAR \leftarrow$ `Sum`($STInfo[part].VAR$)
9 $STInfo[part].CD \leftarrow$ `Distance`($part.center$, $PList.center$)
10 $actQ$.`Sort` **by** $-MVAR$, $-CD$, AFT, $-SVAR$ **in** $STInfo$
11 **return** $actQ$, $STInfo$

B. Dynamic Reactivation-Driven Parallel Disassembly Planning

To meet the demands of real-time, multi-user collaborative VR, we design a parallel disassembly planning algorithm. Its core objective is to identify sets of components that can be removed concurrently. However, parallel attempts introduce a key challenge: disassembling one part can affect the feasibility of removing another simultaneously. Our novel dynamic reactivation strategy is explicitly designed to address these emergent, state-dependent interferences.

The inputs to our algorithm are the model part list $PList$ and the maximum parallelism number $PalNum$. The outputs are the generated parallel disassembly sequence $disSeq$ and the suggested collaboration user count $userCnt$. The complete procedure is detailed in Algorithm 2.

The algorithm operates iteratively. To ensure computational feasibility and robust termination, each iteration begins with a `TDEL` (Time or Depth Exceeded Limit) check (lines 3-4). This mechanism monitors the execution state and halts the process if the accumulated computation time or search depth surpasses predefined thresholds. These thresholds are configurable parameters, allowing users to tailor the planner's termination logic according to their specific time constraints and available computational resources. For the comparative experiments detailed in Section IV, we enforced identical time and depth limits for both our DPDT method and the baseline to ensure a strictly fair evaluation and effectiveness for VR interaction.

a) Preparation and Simulation: In each round, the system first prepares the candidate parts. The active queue $actQ$ is sorted by the `STAwPri` function to prioritize parts with the highest success probability (line 5). The top $PalNum$ candidates are then moved to a temporary parallel queue $PalQ$ (line 6). We employ the DiffRedMax [24] physics engine to simulate the removal of all parts in $PalQ$ simultaneously, strictly enforcing collision detection (lines 7-8). Once the simulation concludes, the system aggregates the successful parts $sucPs$ and updates the suggested user count $userCnt$ based on the current parallel throughput (lines 9-10).

b) Dynamic Reactivation and Dependency Resolution: The core innovation of our method lies in how it processes these simulation results. We introduce a dynamic reactivation strategy that classifies and resolves inter-part dependencies into two distinct categories:

- **Inter-group Reactivation (lines 12-16):** This mechanism addresses dependencies between the currently processing group and the waiting parts. When a part is successfully removed, the algorithm scans the inactive queue $inactQ$ for any components that were previously blocked by this specific part. If such a dependency is resolved, the blocked part is reactivated by promoting it from the inactive queue back to the $actQ$. This effectively unlocks new branches of the search space for subsequent rounds.
- **In-group Reactivation (lines 17-20):** This mechanism handles conflicts within the current parallel batch. If a part fails its disassembly attempt, the system checks if the failure was caused by a collision with another part that was successfully removed in the same parallel round $sucPs$. Since the obstruction is removed simultaneously, this failure is treated as a temporary state conflict rather than a true geometric constraint. The part is reactivated immediately, retaining its active status for reconsideration in next round without increasing attempt failed times AFT.

If neither reactivation condition is met, the attempt is defined as a failure (lines 21-23). The part is penalized by incrementing its AFT, moved to $inactQ$, and removed from $actQ$ to prevent redundant checks.

c) Round Finalization (lines 24-26): Finally, to maintain high parallelism in subsequent rounds, if the part count in $actQ$ falls below the target $PalNum$, the system forcibly replenishes

it by retrieving parts from the inactive queue. The round counter is then incremented, and the loop continues until the assembly is fully processed.

Algorithm 2: Dynamic Reactivation-Driven Parallel Disassembly Planning

```
1  Input: Model part list PList, parallelism number PalNum
2  Output: Model disassembly boolean status, suggest collaboration user count userCnt actQ ← PList while PList.Count() > 0 do
3      if TDEL() = True then
4          return False
5      actQ, STInfo ← STAwPri(actQ)
6      PalQ ← actQ.DeQ(PalNum) results.Clear()
7      for part in PalQ do
8          results[part] ← AtmptDis(PList, part, STInfo[part].VAR, STInfo[part].AFT)
9      sucPs ← results(status = Success)
10     userCnt ← ⌊(userCnt * disRound + sucPs.Count())/(disRound + 1)⌋
11     for part, status, path, contPs in results do
12         if status = Success then
13             PList.Remove(part)
14             disSeq.EnQ(part, path, disRound) for inactP in inactQ do
15                 if STInfo[inactP].contPs.Contain(part) then
16                     inactQ.Remove(inactP) actQ.EnQ(inactP)
17         else
18             STInfo[part].contPs ← contPs
19             if contPs.Contain(any p in sucPs) then
20                 continue
21             else
22                 STInfo[part].AFT += 1
23                 inactQ.EnQ(part) actQ.Remove(part)
24     if actQ.Count() < PalNum then
25         actQ.EnQ(inactQ.DeQ(PalNum-actQ.Count()))
26     disRound += 1
27 return True
```

C. Proximity-Aware Task Assignment

Once a parallel disassembly plan is generated—consisting of parts grouped into sequential rounds—the final step is to distribute these tasks to users in the collaborative VR environment. The goal of our assignment is not only to execute the plan correctly but to do so in a way that minimizes physical effort and maximizes team efficiency. This process comprises proximity calculation, task assignment, and guidance via visual cues.

1) Proximity Calculation and Assignment: The input of our proximity calculation and assignment (Algorithm 3) consists of the planned disassembly sequence *disSeq* and the user count *userCnt*. The output is the set of task assignments *tskAssign*.

The execution flow begins by initializing the user list and the assignment container (line 3). The algorithm then enters a main loop that persists while parts remain in the sequence (line 4). In each iteration, the system first identifies the subset of users who are currently available (line 5); if no users are available, the process waits for the next cycle (lines 6-7). Concurrently, it determines the set of viable tasks for the current phase by retrieving parts associated with the minimum remaining round ID (lines 8-12). With these groups identified, the optimal pairing is computed using the distance minimization function (line 15). Finally, the system updates the user states and removes the assigned parts from the sequence (lines 16-19) before returning the complete assignment list (line 20).

To quantify the cost of assigning a user to a task, we calculate the 3D Manhattan distance (L_1 norm) between the user's virtual position and the centroid of the target part. While Euclidean distance represents the shortest path in free space, it implies the ability to move directly through obstacles. In a realistic disassembly scenario, the assembly object typically occupies the center of the workspace, requiring users to physically navigate around it rather than reaching through it. Furthermore, our VR interaction design incorporates vertical movement via controller inputs as detailed in Section V-A, making vertical travel a distinct component of user effort. The 3D Manhattan distance, calculated as $|dx|+|dy|+|dz|$, effectively penalizes diagonal paths that would intersect the central workspace, thus serving as a more realistic proxy for the actual navigational effort and vertical reaching workload required in this constrained environment.

The `AssignMinManhatDis` function solves for the set of pairings that minimizes the total sum of these distances. Formally, let U_{avail} be the set of available users and P_{next} be the set of candidate parts for the current round. We seek an optimal assignment mapping $\mathscr{M} : U' \to P'$, where $U' \subseteq U_{avail}, P' \subseteq P_{next}$ that satisfies:

$$\min_{\mathscr{M}} \sum_{(u,p)\in\mathscr{M}} (|x_u - x_p| + |y_u - y_p| + |z_u - z_p|) \tag{2}$$

where (x_u, y_u, z_u) and (x_p, y_p, z_p) represent the 3D coordinates of the user and the part centroid, respectively. This objective function corresponds to the classic Linear Assignment Problem (LAP). In our implementation, we utilize the Hungarian algorithm to solve Eq. (2), ensuring a globally optimal matching in polynomial time for each round.

It is essential to note that while Algorithm 3 describes the assignment logic as a sequential procedure for clarity, our implementation in the VR system operates in an event-driven manner. The assignment routine is triggered in real-time whenever a user completes a task or a new parallel round becomes accessible. This ensures that the system responds dynamically to the asynchronous progress of multiple users, rather than adhering to a rigid, pre-calculated timeline.

Furthermore, this proximity-aware optimization is particu-

Algorithm 3: Proximity Calculation and Assignment

Input: Planned disassemble sequence *disSeq*, user count *userCnt*
Output: Task Assignments *tskAssign*
Initialize *userCnt* users as *userL*, *tskAssign* ← `new List()`
while *disSeq*.`Count()` > 0 **do**
  avilU ← *userL* where *user.task* is `null`
  if *avilU*.`Count()` = 0 **then**
    continue
  minR ← `Min`(*disP.disRound* in *disSeq*)
  avilP ← `new Queue()`
  for *disP in disSeq* **do**
    if *disP.disRound* = *minR* **then**
      avilP.`EnQ`(*disP*)
  if *avilP*.`Count()` = 0 **then**
    break
  tskAssignRound ← `AssignMinManhatDis` (*avilU*, *avilP*)
  for *(user, part)* ***in*** *tskAssignRound* **do**
    user.task ← *part*
    disSeq.`Remove`(*part*)
  tskAssign.`AddAll`(*tskAssignRound*)
return *tskAssign*

larly effective when the *PalNum* exceeds the number of available users. In such scenarios, the algorithm selects the optimal subset of tasks that minimizes the immediate aggregate travel cost for the current users, effectively creating a dynamic buffer of valid tasks. This mechanism allows the system to maintain workflow continuity and optimize physical efforts even when the workforce is limited.

2) Assignment Guidance in VR: Effective task assignment must be communicated to the users clearly and intuitively. Once the optimal pairings are found, the system updates each user's task and removes the part from the sequence queue. Concurrently, it provides two synergistic forms of visual guidance in the VR interface, as illustrated in Fig. 3:

- **Target Highlighting:** As shown in the top row of Fig. 3, assigned parts are visually highlighted with unique colors corresponding to specific users. This allows for immediate and unambiguous target identification among multiple collaborators.
- **Path Visualization:** As illustrated in the lower images of Fig. 3, the system offers an optional guidance feature that visualizes removal paths computed in the physical simulation, using dotted trajectory lines and ghosted target positions.

Together, these components create a seamless and ergonomic workflow, translating an optimized disassembly plan into efficient and intuitive collaborative action.

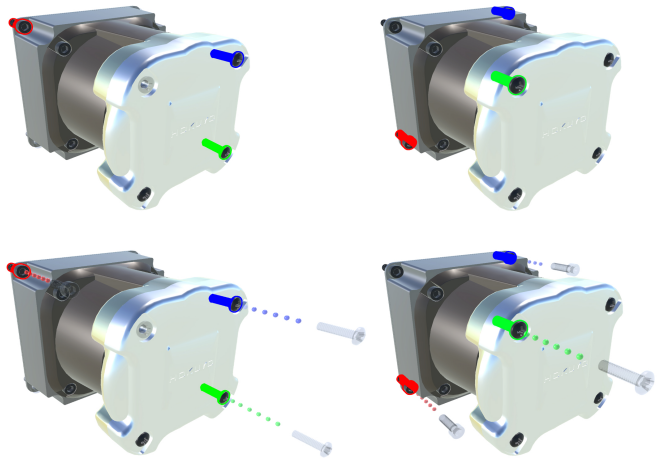


Fig. 3: Visual guidance workflow for multi-user collaborative disassembly across two consecutive rounds. The upper images demonstrate target highlighting for three users to identify active parts, while the lower images reveal the corresponding disassembly trajectories using dotted lines and ghosted final positions to guide users along collision-free paths.

IV. Experiment

A. Experimental Setup

All planning algorithm experiments were conducted on a workstation equipped with an Intel Core i7-13700K CPU and 64 GB of RAM. The planning computations were performed exclusively on the CPU, as our framework is designed for efficiency on general-purpose hardware without requiring GPU acceleration.

To evaluate the effectiveness of our DPDT framework, particularly its planning component, we conducted experiments using the large-scale dataset publicly released with the Assemble Them All (ATA) paper [5]. This dataset consists of numerous 3D models geometrically pre-processed to be watertight and normalized.

To ensure a rigorous and fair comparison, our simulation environment aligns strictly with the baseline method ATA [5]. We utilize **DiffRedMax** [24], a differentiable simulator known for its accurate penalty-based contact model and implicit time integration. By adopting the exact same physics engine and collision detection thresholds as the baseline, we eliminate environmental variables and focus the evaluation purely on planning efficiency.

To ensure practical applicability, we established a time limit of 3600 seconds for the entire model disassembly process and a limit of 120 seconds for disassembling a single part. Furthermore, both ATA and DPDT employ the Breadth-First Search (BFS) algorithm. To mitigate issues arising from part interpenetration, we set the maximum BFS search depth to 15. This constraint helps filter out models with severe initial overlaps, focusing our evaluation on structurally plausible assemblies.

Under these defined limitations, we selected a subset of 3671 models from the original dataset. Within this subset, 382

models consisted of 20 or more individual parts, referred to as the *complex subset*.

B. Methods Compared and Metrics

We compared our DPDT approach, which uses Algorithm 2 as its planning core, against the baseline method, Assemble Them All (ATA) [5]. We selected ATA as the primary baseline because it provides a standard, physics-based foundation for disassembly feasibility validation that is structurally comparable to our simulation environment. While newer methods such as ASAP [8] have been proposed, their focus on robotic grasping and stability constraints results in computational latencies that are incompatible with interactive VR applications. Consequently, ATA remains the most suitable and efficient benchmark.

Both methods were evaluated with parallel disassembly numbers $P_n \in \{1,2,3,4,5\}$, indicating the maximum parts attempted simultaneously per planning round. The parallelism number, P_n, is treated as a key hyperparameter that can be tuned based on assembly complexity and available computational resources; for this study, we systematically varied its value to evaluate its impact. While both methods operate sequentially for $P_n = 1$, DPDT uniquely employs our dynamic contact-based reactivation strategy when $P_n > 1$ to manage dependencies arising during parallel attempts. This dynamic update mechanism distinguishes DPDT from the baseline ATA under parallel execution conditions where $P_n > 1$.

We used the following metrics for comparison:

- **Average Disassembly Plan Time**: Calculated as the total time taken to generate a full disassembly plan for a model, averaged across all models in the respective test set. A lower AvgT indicates higher efficiency.
- **Disassembly Plan Success Rate**: This metric represents the proportion of individual disassembly steps, predicted as valid by the algorithm and subsequently tested in a physical simulation, that are successfully executed without collision or invalid states, averaged over all steps and models. A higher SR indicates better prediction quality and plan reliability.

Considering the computational demands and the relevance of parallelism for complex scenarios, methods with $P_n = 1$ were evaluated on the full set of 3671 models. Configurations with $P_n \in \{2,3,4,5\}$ were primarily evaluated on the subset of 382 complex models, which are those with more than 20 parts. For comprehensive comparison, results for $P_n = 1$ on this complex subset are also included.

C. Results and Analysis

The experimental results are presented and analyzed below, focusing first on the quality and reliability of the generated plans, and second on the computational efficiency.

1) Plan Quality and Reliability: Analysis of the Disassembly Plan Success Rate (SR) demonstrates the superior reliability of our DPDT framework. DPDT consistently outperforms the baseline ATA across varying degrees of parallelism (P_n) and model complexities. The results are summarized in Table I, and the trend with model complexity is shown in Fig. 4.

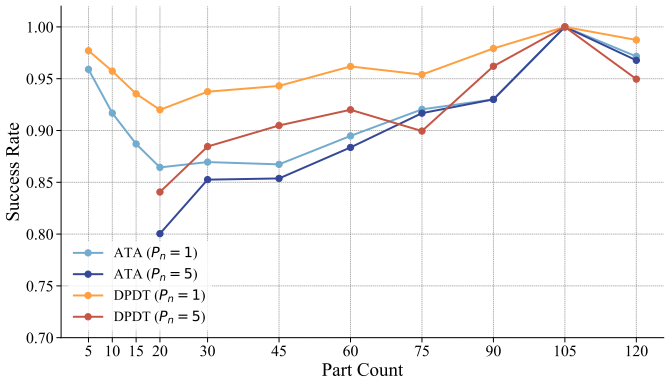


Fig. 4: Scalability analysis of Disassembly Plan Success Rate (SR) with respect to assembly complexity. The trend lines demonstrate the robustness of DPDT compared to the baseline as the structural complexity increases.

In the sequential setting ($P_n = 1$), DPDT achieves a Success Rate (SR) of 93.64% on the full dataset, which represents a significant improvement over the 86.90% rate achieved by ATA. This margin widens to 9.28 percentage points on the complex subset with > 20 models, validating the robustness of our spatial-temporal prioritization strategy in handling intricate geometries. As parallelism increases, the complexity of inter-part dependencies grows. While the SR for both methods naturally declines under higher P_n values, DPDT exhibits greater resilience. At $P_n = 5$, DPDT maintains a high reliability of 87.47%, whereas the baseline performance drops to 82.21%.

The scalability analysis in Fig. 4 reinforces this finding, showing that DPDT method consistently achieves higher reliability across the spectrum of part counts compared to the baseline. The parallel DPDT method maintains a high level of reliability, significantly surpassing both versions of the baseline. This demonstrates the robustness of DPDT and the effectiveness of its dynamic update strategy in handling the increased complexity and potential conflicts arising from parallel disassembly attempts.

2) Planning Efficiency: In terms of computational efficiency, DPDT demonstrates a distinct advantage, particularly for complex assemblies where search space explosion is a critical bottleneck. Table II and Fig. 5 summarize these findings.

For sequential planning ($P_n = 1$) on the complex subset, DPDT reduces the average planning time from 41.37s to 16.36s, achieving a speedup factor of approximately 2.5 times. The efficiency gains become most pronounced under parallel configurations. With $P_n = 5$, DPDT achieves an average time of 8.66s, which corresponds to a 4.7 times speedup compared to the baseline's sequential performance, and is over 2.1 times faster than ATA's best parallel performance of 18.40s.

The time scalability analysis in Fig. 5 provides deeper insights. As expected, planning time generally increases with the number of parts. However, our parallel DPDT method ($P_n = 5$) demonstrates superior time scalability, typically yielding the fastest results for more complex models with higher part counts.

In summary, the comparative analysis confirms the superior

TABLE I: Disassembly Plan Success Rate Comparison (%).

Method	3671 models	382 models (>20 parts)				
	$P_n = 1$	$P_n = 1$	$P_n = 2$	$P_n = 3$	$P_n = 4$	$P_n = 5$
ATA	86.90	84.42	83.78	83.25	82.65	82.21
DPDT	**93.64**	**93.70**	**89.91**	**88.55**	**87.92**	**87.47**

TABLE II: Average Disassembly Plan Time Comparison (s).

Method	3671 models	382 models (>20 parts)				
	$P_n = 1$	$P_n = 1$	$P_n = 2$	$P_n = 3$	$P_n = 4$	$P_n = 5$
ATA	6.700	41.367	29.758	24.135	19.343	18.400
DPDT	**2.856**	**16.358**	**14.512**	**15.905**	**10.095**	**8.657**

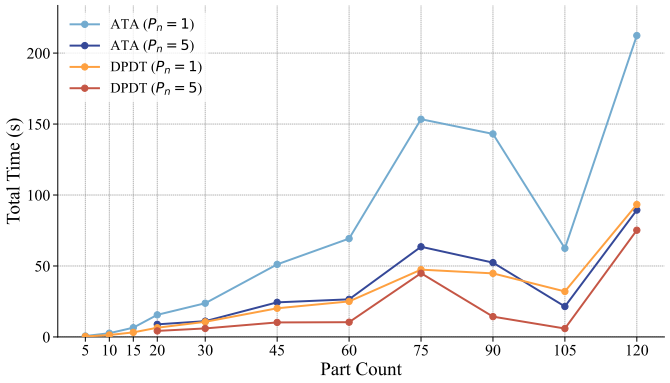


Fig. 5: Scalability analysis of Average Disassembly Plan Time (AvgT) with respect to assembly complexity. The chart illustrates the computational efficiency and time-scaling advantage of DPDT for handling complex models compared to the baseline.

performance and robustness of DPDT. This validates that our spatial-temporal aware prioritization strategy is highly effective at guiding the planner to find viable disassembly solutions more quickly and reliably.

D. Ablation Studies

To validate our specific design choices and understand the individual contributions of DPDT's key components, we conducted two targeted ablation studies.

1) Contribution of Prioritization Features: To evaluate the contribution of each feature in DPDT's spatial-temporal aware prioritization strategy, as defined in Algorithm 1 (line 11), we tested variants of our full method (MCAS) by removing or reordering its component features. Our full method prioritizes parts using the sorting key composed of the features $-MVAR$, $-CD$, AFT, $-SVAR$. The tested variants are:

- **MCAS (Ours)**, **MCSA**, **MSCA**: Reordering the features.
- **CAS**, **AS**, **S**: Sequentially removing features.

The results of these ablation studies are presented in Table III. The ablation results provide valuable insights into the contribution of the different sorting features and their specific ordering within DPDT's part selection strategy. Our full DPDT (MCAS) method consistently achieves the best overall performance.

Comparing DPDT (MCAS) with variants MCSA and MSCA, which use the same features but in different orders, reveals the importance of the chosen prioritization. While MCSA performs similarly to DPDT (MCAS) in terms of sequential success rate, it is slightly slower. MSCA shows a more noticeable drop in performance, particularly in success rate. In the parallel setting, DPDT (MCAS) demonstrates clearer advantages over both MCSA and MSCA in both speed and reliability. This suggests that prioritizing MVAR first, followed by CD, and then considering AFT before SVAR, provides the most effective guidance for the disassembly search.

The experiments involving the removal of features underscore the importance of the selected features. Removing MVAR, as in variant CAS, leads to a significant increase in planning time and a notable decrease in success rate compared to DPDT (MCAS), especially under parallel execution. This confirms that the Maximum Visibility Area Ratio is a critical feature for identifying promising candidate parts for removal. Further removing geometric information such as CD (variant AS), or relying solely on SVAR (variant S), results in additional performance degradation, yielding the longest planning times and the lowest success rates. This demonstrates that the synergistic combination of features representing visibility, geometry, and attempt history is crucial for efficient and reliable disassembly planning.

In conclusion, our ablation studies validate the sorting strategy ($-MVAR$, $-CD$, AFT, $-SVAR$) for DPDT as the most effective combination and ordering among the tested variants. MVAR emerges as a particularly influential criterion, and the combined use of visibility, geometry, and attempt history features is essential for achieving the high efficiency and reliability demonstrated by DPDT.

2) Effectiveness of the Dynamic Reactivation Mechanism: To specifically isolate and validate the contribution of our dynamic reactivation mechanism, as outlined in our second innovation, we conducted an additional ablation study. We created a variant of DPDT where this mechanism was disabled: upon a successful disassembly, the planner no longer intelligently queried the inactive queue to reactivate previously

TABLE III: Ablation study results for different sorting strategy variants.

Strategy Variant	Sequential Disassembly ($P_n = 1$)		Parallel Disassembly ($P_n = 5$)	
	Average Time (s)	Success Rate (%)	Average Time (s)	Success Rate (%)
MCAS (Ours)	**16.358**	**93.70**	**8.657**	**87.47**
MCSA	17.626	93.69	8.981	87.12
MSCA	18.598	91.10	9.670	86.21
CAS	23.172	90.30	10.612	84.50
AS	23.638	88.69	12.016	83.68
S	23.514	88.88	11.806	83.80

blocked parts. Both the full DPDT method and this ablated version were tested on the complex subset with a parallelism setting of $P_n = 5$. We specifically selected this high-parallelism configuration because it induces the highest density of concurrent disassembly attempts. By maximizing the likelihood of inter-part dependencies and temporary blockages, this setting serves as the most rigorous test for validating the necessity and efficacy of our dynamic reactivation strategy.

The full DPDT framework demonstrated its superiority by achieving an average plan time of 8.657s over the 9.863s required by the version without dynamic reactivation. Similarly, its success rate reached 87.47%, outperforming the ablated version's 86.11%. The 12.2% reduction in average time stems from the mechanism's ability to prevent the planner from redundant re-evaluations of parts that are known to be blocked, instead allowing it to more directly explore newly viable candidates. This leads to a more direct and efficient search path. Similarly, the 1.36% point improvement in success rate enhances the planner's robustness. It allows the algorithm to correctly identify and capitalize on new disassembly opportunities that emerge within a single parallel step—scenarios where a less adaptive planner might fail. This is particularly critical in densely packed assemblies where dependencies are complex.

These results confirm that the dynamic reactivation mechanism is a vital and effective component of our parallel planning strategy, crucial for resolving the complex dependencies that arise during concurrent disassembly attempts.

TABLE IV: Ablation study results for the Dynamic Reactivation mechanism on the complex subset ($P_n = 5$).

Method Variant	Average Time (s)	Success Rate (%)
DPDT (Full)	**8.657**	**87.47**
DPDT (Ablated)	9.863	86.11

V. User Study

To evaluate the physical effort and collaborative efficiency of our DPDT framework in a VR environment, we conducted a user study. This study compared DPDT against the baseline ATA [5], focusing on user performance and physical effort during multi-part disassembly tasks of varying complexity and required collaborator counts.

A. Participants and Experimental Design

a) Participants and Apparatus: We recruited 12 participants, comprising 8 males and 4 females aged between 20 and 28, from our university. All participants were right-handed and had normal or corrected-to-normal vision. The study was conducted using Pico 4 Pro VR headsets within a collaborative virtual environment developed in Unity (version 2022.3.5f1c1). The simulation ran on a high-performance PC (Intel Core i7-13700K, NVIDIA GeForce RTX 4090) to ensure real-time synchronization and a smooth, high-fidelity experience.

b) Task Description and Interaction: The fundamental task required users to disassemble parts from a target object in a specific sequence. To ensure experimental consistency across all trials, all participants started from the world origin (0, 0, 0), and the virtual objects were positioned at standardized locations relative to the users, as shown in Fig. 6.

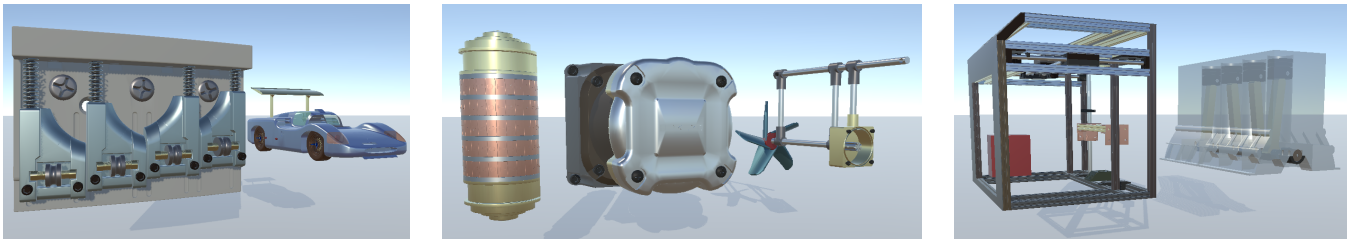
The interaction workflow for each part consisted of three steps:

1) **Identification:** Upon entering the virtual environment, users identified their assigned target part, which was indicated by a visual highlight (a colored glowing outline) and a path trajectory line.
2) **Navigation:** Users navigated from the origin toward the target object using joystick-based locomotion. Vertical height adjustments were made using discrete controller buttons, specifically designated as X or A for descending and Y or B for ascending.
3) **Disassembly:** To focus the evaluation on sequence planning and navigation efficiency rather than physical manipulation difficulty, we simplified the interaction method. Users were required to reach out and touch the target part with their virtual controller and press the trigger button.

Successful interactions caused the part to disappear, advancing the sequence. Interactions with incorrect parts were ignored.

c) Experimental Design: We employed a within-subjects design with two independent variables: method, comparing DPDT against the ATA baseline, and scenario, which encompassed four distinct configurations (S1P1, S2P2, S3P3, and S2P5). The scenarios varied in object complexity and the number of collaborators:

- **S1P1 (1 User):** 60 parts, $P_n = 1$.
- **S2P2 (2 Users):** 66 parts, $P_n = 2$.
- **S3P3 (3 Users):** 126 parts, $P_n = 3$.



(a) Scenario 1: 2 Objects (60 parts) (b) Scenario 2: 3 Objects (66 parts) (c) Scenario 3: 2 Objects (126 parts)

Fig. 6: Objects used in the user study scenarios.

- **S2P5 (2 Users):** 66 parts, $P_n = 5$, testing high-parallelism planning with fewer users.

For the baseline (ATA) condition, we utilized the same parallelism setting P_n as the experimental condition. However, since the original ATA framework lacks a multi-user task allocation protocol, we implemented a standard *Round-Robin* strategy. In this baseline approach, available parallel tasks are assigned to users in a fixed cyclic order.

d) Procedure: The 12 participants were organized into groups according to the scenario requirements (12 groups for S1, 6 groups for S2, and 4 groups for S3). The study followed a structured protocol:

1) **Introduction and Training:** Participants were briefed on the task and given a 5-minute practice session with a tutorial object to familiarize themselves with the VR locomotion and disassembly interaction mechanisms.
2) **Formal Trials:** Each group performed the disassembly tasks for the assigned scenarios using both methods. To eliminate order effects, the sequence of methods (DPDT then ATA, or ATA then DPDT) was counterbalanced using a Balanced Latin Square design across groups.
3) **Timing and Breaks:** Based on preliminary tests, each disassembly session lasted approximately 5 to 15 minutes depending on the scenario complexity and method used. To prevent fatigue and motion sickness, a mandatory 5-minute break was enforced between the two conditions. The total duration of the experiment was approximately 40 minutes per group.

B. Metrics

We collected quantitative metrics to assess performance and user effort:

- **Time per Part (T/Part, s)**: Total completion time divided by the number of parts.
- **Head Movement (HeadM, m)**: Total distance traveled by the user's head-mounted display.
- **Hand Movement (HandM, m)**: Average total distance traveled by the left and right hand controllers.
- **Head Rotation (HeadR, °)**: Total rotation of the user's head.
- **Hand Rotation (HandR, °)**: Average total rotation of the left and right hand controllers.

C. Results and Analysis

We analyzed the collected metrics using paired-samples t-tests to compare DPDT against ATA within each scenario. The detailed results for task efficiency are presented in Table V, while the outcomes for physical effort are visualized in Fig. 7. The findings show a consistent and statistically significant improvement with DPDT over the baseline in most of the cases.

1) Task Efficiency: As detailed in Table V, DPDT significantly reduced the Time per Part (T/Part) across all four experimental scenarios, with statistical significance established at the $p < .001$ level. In the low-complexity collaborative scenario (S2P2), users utilizing DPDT completed tasks 56.5% faster than with the baseline, reducing the average time per part from 8.40s to 3.65s. Even in the most complex scenario (S3P3, 126 parts), DPDT yielded a 37.5% reduction in task time, lowering it from 4.48s to 2.80s. Notably, comparing the high-parallelism condition S2P5 ($P_n = 5$) with S2P2 ($P_n = 2$) reveals that increasing algorithmic parallelism beyond the user count further optimized the workflow, dropping the time from 3.65s to 2.77s. This suggests that our task assignment algorithm effectively utilizes the buffer of available parallel tasks to minimize idle time.

TABLE V: Task Efficiency Results: Time per Part (s)

Scenario	Method	Mean ± SD (s)	p-value	Effect Size
S1P1	ATA	8.00 ± 0.50	$< .001$	Very Large
	DPDT	$\mathbf{4.79 \pm 0.32}$		
S2P2	ATA	8.40 ± 0.43	$< .001$	Very Large
	DPDT	$\mathbf{3.65 \pm 0.24}$		
S3P3	ATA	4.48 ± 0.33	$< .001$	Very Large
	DPDT	$\mathbf{2.80 \pm 0.19}$		
S2P5	ATA	5.52 ± 0.41	$< .001$	Very Large
	DPDT	$\mathbf{2.77 \pm 0.24}$		

2) Physical Effort: As both methods operate within the same VR system and utilize identical interaction logic, we used the following objective metrics to quantify physical effort.

DPDT significantly minimized user movement. In the highly complex S3P3 scenario, total head movement was reduced by 48.2%, dropping from 137.25m to 71.15m ($p < .001$). Similarly, hand movement distance saw reductions across all

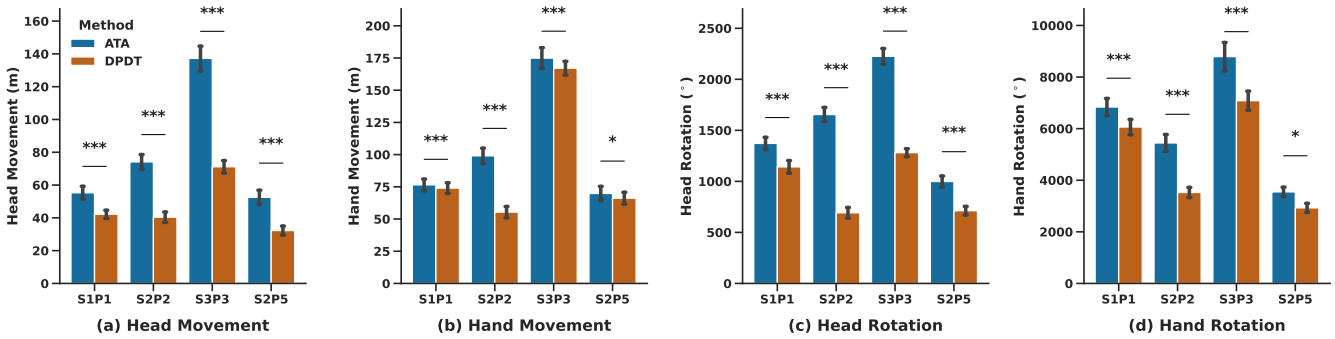


Fig. 7: User study results comparing DPDT and ATA across four conditions. For each metric, lower values generally indicate better performance. Error bars represent standard deviation. The asterisks (***) denote a statistically significant difference with $p < .001$.

scenarios, reaching a peak reduction of 44.1% in S2P2, where the distance fell from 99.03m to 55.37m ($p < .001$). The system also reduced the need for physical reorientation. Head rotation decreased by up to 58.2% in the S2P2 condition ($p < .001$), and hand rotation decreased by 35.3%. These findings statistically validate that the proximity-aware task assignment strategy successfully translates algorithmic efficiency into a tangible reduction in user physical workload.

3) Scenario-Specific Observations: While DPDT outperformed ATA in all conditions, the margin of improvement varied. The time improvement in the complex S3P3 scenario was 37.5%, which was less pronounced than the 56.5% improvement observed in the S2P2 scenario. This may be because for highly complex objects, the primary bottleneck can shift from algorithmic planning to human operational speed. Nevertheless, the significant reductions in physical effort in S3P3 still highlight the superior ergonomic quality of the DPDT-generated plan.

The comparison between S2P5 and S2P2 is also revealing. Increasing the plan's parallelism ($P_n = 5$) beyond the number of users (2) not only accelerated the task but also further reduced user effort. This confirms that providing more concurrent task options allows the proximity-aware assignment algorithm to make better decisions, optimizing user paths more effectively.

In summary, our user study provides strong quantitative evidence for the practical benefits of the DPDT framework. DPDT enables significantly faster task completion and reduces user physical effort compared to the baseline.

VI. Conclusion

In this paper, we proposed DPDT, a cohesive framework designed to streamline collaborative disassembly in VR through intelligent planning and tasking. The core of DPDT is a novel planning engine guided by a prioritization strategy based on spatial-temporal features, coupled with a parallel planning algorithm featuring a dynamic reactivation mechanism to resolve inter-part dependencies. This engine is integrated with a proximity-aware task assignment strategy tailored for multi-user interaction.

Through extensive experiments, we demonstrated that DPDT's planning algorithm significantly outperforms the state-of-the-art baseline. Specifically, for complex assemblies, DPDT achieved a computational speedup of up to 4.7 times over the baseline, while maintaining a 5.3-9.3 percentage point improvement in plan reliability across all test sets. The effectiveness of our parallel planning was particularly evident for models with high part counts, where computation time was nearly halved on average. Targeted ablation studies further confirmed that the dynamic reactivation mechanism is crucial for this performance gain in parallel scenarios. Furthermore, our user study translated these computational gains into tangible user benefits. In a collaborative VR setting, users achieved a reduction in task completion time ranging from 37.5% to 56.5% across different complexity scenarios, along with a reduction in physical effort of up to 58.2% compared to the baseline. Increasing the plan's parallelism beyond the number of available users further improved efficiency, underscoring the powerful synergy between our efficient parallel planning engine and task assignment strategy.

While our DPDT framework presents significant advancements, certain limitations highlight promising directions for future research. A primary limitation lies within the geometric reasoning of the core planning algorithm. Currently, our method primarily evaluates translational removability based on projection and depth maps. This approach is highly effective for direct pull-out actions but does not explicitly model complex rotational movements, such as navigating non-linear helical paths or unscrewing threads. Future work should aim to extend the planner by integrating 6-DOF motion planning or hybridizing our heuristic approach with sampling-based methods. A key challenge will be addressing these complex rotational constraints while maintaining the high computational efficiency required for real-time interactive VR applications. Furthermore, the assumption of a standard gravitational environment overlooks stability constraints crucial for microgravity applications or handling delicate structures. Integrating stability analysis into DPDT's planning process is a necessary step for ensuring safe and predictable disassembly in such specialized contexts. The efficiency gains from parallel

planning, while substantial, are inherently bound by the task's structure. For highly constrained objects with a strictly sequential disassembly path, the benefits of parallelism diminish, creating bottlenecks. Future research could explore hybrid planning approaches that dynamically switch between parallel and optimized sequential strategies based on the assembly's current state, or develop predictive models to anticipate bottlenecks and optimize task allocation further down the dependency chain. Improvements can also be envisioned for the task assignment strategy. The current algorithm, while effective, is greedy as it optimizes assignments on a round-by-round basis. Future iterations could incorporate more sophisticated 3D path-finding algorithms that account for obstacles, or explore look-ahead strategies that consider future task locations for a more globally optimal assignment. Additionally, the system could be enhanced to consider user-specific factors such as skill level or fatigue dynamically, or even allow users greater agency in task selection from a set of valid options.

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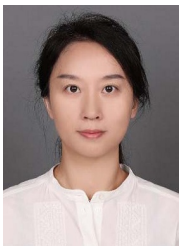
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